

Notes: Atkeson & Kehoe (JPE, 2005)

Modeling and Measuring Organization Capital

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1 Big picture

- **Question.** How large are the payments to organization capital in U.S. manufacturing, and can the plant life cycle be used to measure them directly?
- **Core idea.** Manufacturing plants are born small, grow over many years, and eventually exit. Atkeson and Kehoe interpret this life cycle as evidence of plant-specific organization capital.
- **Measurement strategy.** The paper builds a quantitative industry-evolution model and calibrates plant productivity dynamics to match employment, job creation, and job destruction by plant age.
- **Main result.** With the baseline span-of-control parameter, payments to organization capital are about 3.3% of manufacturing output, approximately 40% of the NIPA-based estimate of total intangible-capital payments in manufacturing.
- **Economic takeaway.** A large part of measured profits in manufacturing can be interpreted as compensation for owners who financed the slow accumulation of plant-specific knowledge.

2 Incremental contribution of the paper

- **From residual intangible capital to structured measurement.** Earlier macro accounting could infer intangible capital only indirectly. This paper provides a disciplined way to measure one component—organization capital—using plant-level life-cycle facts.
- **Bridges industry dynamics and macro income shares.** It links Hopenhayn–Rogerson/Jovanovic style plant dynamics to national-accounting factor shares.
- **Organization capital as owner-owned plant knowledge.** The paper focuses on organization capital embodied in plants and transferable through ownership, distinct from firm-specific human capital embodied in workers.
- **Back-loaded rents as the measurement insight.** Free entry implies zero present value for new plants, but positive aggregate organization rents arise from back-loaded plant life cycles and positive interest rates.
- **Direct calibration to job-flow data.** The learning process is chosen to match U.S. plant-age employment, job creation, and job destruction moments.
- **Quantitative benchmark.** The estimate that organization capital accounts for roughly 40% of intangible-capital payments provides a benchmark for macro, growth, and firm-dynamics research.

3 Model objects and measurement

3.1 Plant life cycle

Plants enter at age zero with frontier blueprints but little accumulated operating knowledge. Plants grow as their specific productivity evolves stochastically with age. Plants can exit; if a plant shuts down, its organization capital is lost.

Interpretation: The life cycle is the observable shadow of organization-capital accumulation.

3.2 Organization capital and organization rents

Organization capital is the pair (A, s) : specific productivity A and age s . Organization rents are variable profits minus the fixed operating/managerial cost. Young plants generally have low rents; older successful plants have higher rents.

Interpretation: The measured payments to organization capital come from the cross-section of plants of different ages.

4 Models and theoretical specifications

4.1 Simplified plant life-cycle model

A plant of age s hires labor l and produces

$$y_s = f_s(l_s).$$

Variable profits are

$$d_s = \max_l \{f_s(l) - wl\}.$$

Free entry implies

$$\sum_{s=0}^N \left(\frac{1}{1+i} \right)^s (d_s - w_m) = 0.$$

Aggregate payments to organization capital are

$$p = \sum_{s=0}^N (d_s - w_m).$$

Interpretation: If rents are back-loaded and the real interest rate is positive, $p > 0$ even though the value of a newborn plant is zero.

4.2 Full quantitative model

A plant with organization capital (A, s) produces

$$y_t = z_t A F(k_t, l_t)^\nu.$$

Static variable profits are

$$d_t(A) = \max_{k,l} \{z_t A F(k, l)^\nu - r_t k - w_t l\}.$$

The value of a plant satisfies a Bellman equation of the form

$$V_t(A, s) = \max\{0, d_t(A) - w_{mt} + \text{discounted } E[V_{t+1}(A\varepsilon, s+1)]\}.$$

Interpretation: The state variables (A, s) summarize plant-specific knowledge and survival/growth prospects.

5 Calibration, measurement, and quantitative design

The paper uses macro parameters from McGrattan and Prescott and calibrates the productivity-shock process to match U.S. manufacturing plant-age moments: employment shares, job creation, job destruction, failing-plant destruction, and growth-bin distributions. The physical capital share is matched to manufacturing data. The baseline span-of-control parameter is $\nu = 0.85$.

Interpretation: The paper is calibrating a structural plant-life-cycle model to reproduce industry-evolution facts and then reading off implied organization rents.

6 Section-by-section study guide

- **Introduction.** Sets up the measurement problem and the finding that organization capital is large.
- **Section I.** Shows why free entry is consistent with positive aggregate organization rents.
- **Section II.** Builds the full model with specific productivity, age, entry, exit, capital, labor, and managers.
- **Section III.** Calibrates macro parameters and plant productivity dynamics.
- **Section IV.** Shows the model fit to employment and job-flow moments.
- **Section V.** Reports output shares and organization-capital payments.

7 Tables and figures

7.1 Figure 1: Employment statistics by manufacturing plant age in the model and U.S. data

Read: The model and data both place more than half of employment in plants older than 25 years; overall creation and destruction rates in the model are about 10.2% versus roughly 8.3–8.4% in the data.

Detailed interpretation: This is the main calibration validation figure. It checks whether the model matches the joint facts that plants start small, surviving old plants are large, and job flows remain active across ages.

Economic message: The model can match the core plant life cycle, supporting its use to infer organization rents.

7.2 Figure 2: Means and standard deviations of shocks to plant size by age of plant

Read: The mean shock starts positive for young plants and becomes negative for older plants; the standard deviation is roughly constant.

Detailed interpretation: The calibrated process lets young surviving plants grow rapidly while preserving substantial idiosyncratic risk at all ages.

Economic message: Organization capital accumulation is prolonged but risky.

7.3 Figure 3: Job destruction in failing plants by age of plant

Read: Old failing plants account for the largest destruction shares; the model captures the broad pattern but overstates some young-plant failure destruction.

Detailed interpretation: This tests whether the model can capture destruction of accumulated organization capital through exit.

Economic message: Organization capital is not riskless: plant-specific knowledge can be destroyed by failure.

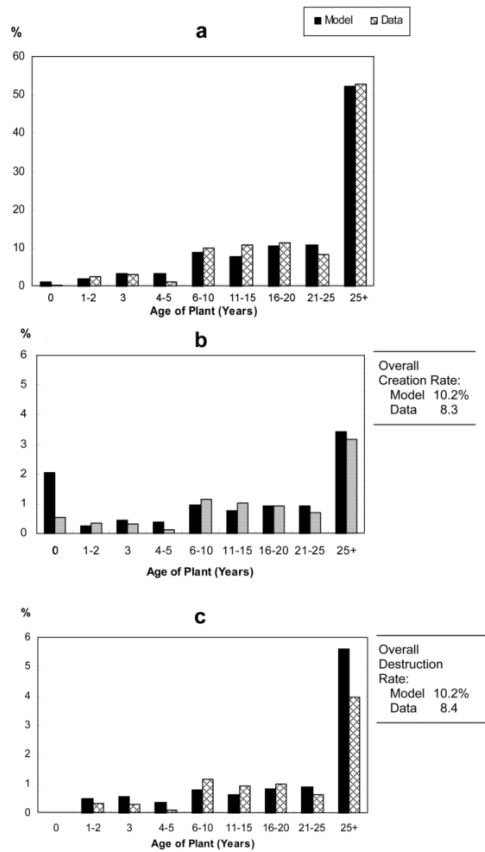


Figure 1: employment, job creation, and job destruction by plant age

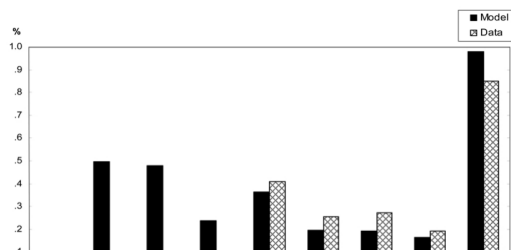


Figure 3: failing-plant job destruction by age

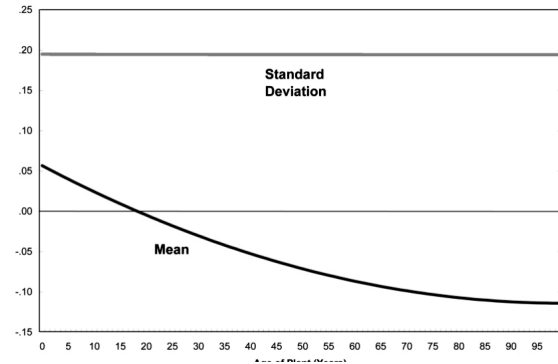


Figure 2: calibrated age-dependent productivity-shock process

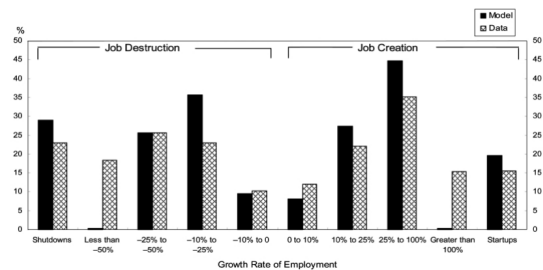


FIG. 4.—Distribution of job creation and destruction (data from Davis et al. 1996)

Figure 4: distribution of job creation and destruction

7.4 Figure 4: Distribution of job creation and destruction

Read: The figure compares job creation and destruction across growth-rate bins, including startups and shutdowns.

Detailed interpretation: The model is evaluated not just on average growth but on the distribution of expansions and contractions.

Economic message: Gross reallocation is central to measuring organization-capital accumulation and destruction.

7.5 Figure 5: Average productivity of plants by age in U.S. data

Read: Average productivity shows no strong systematic age trend.

Detailed interpretation: This helps explain why plant growth can reveal organization capital even when conventional productivity averages are flat by age.

Economic message: Organization capital is reflected in plant size and rents, not necessarily in simple age-productivity averages.

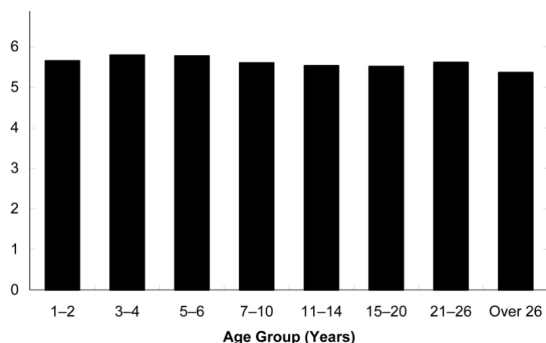


Figure 5: average productivity by plant age

	DATA ON U.S. MANUFACTURING*	MODEL		
		$\nu = .80$	$\nu = .85$	$\nu = .90$
		Shares of Output (%)		
Labor	72.2	75.7	76.8	77.9
Workers	. . .	60.1	65.1	70.1
Managers	. . .	15.6	11.7	7.8
Physical capital	19.9	19.9	19.9	19.9
Intangible capital	8.0	. . .	. . .	. . .
Organization capital	. . .	4.4	3.3	2.2
Other Payments				
Net payments to physical capital [†]	7.3	8.9	8.9	8.9

* U.S. manufacturing data are described in App. A.

[†] Net payments to physical capital are measured net of new investment, i.e., $rk - x$.

Table 1: accounting for output in U.S. manufacturing

7.6 Table 1: Accounting for output in the U.S. manufacturing sector

Read: In the baseline model with $\nu = 0.85$, organization capital receives 3.3% of output. U.S. manufacturing data imply 8.0% of output paid to intangible capital.

Detailed interpretation: The baseline estimate accounts for roughly 40% of measured intangible-capital payments. Organization-capital payments are also more than one-third the size of net physical-capital payments.

Economic message: Organization capital is a large, measurable component of manufacturing income.

8 Result map

- **Plant life-cycle facts discipline organization capital.** Employment and job-flow moments reveal the evolution of hidden plant-specific productivity.
- **Free entry does not imply zero aggregate rents.** New plants have zero value, but older plants earn positive rents.
- **Positive interest rates make waiting costly.** Organization rents compensate owners for financing early low-rent years.
- **The calibrated learning process is long-lived.** Plant cohorts grow for many years.
- **Organization capital is large.** The baseline estimate is about 3.3% of output, roughly 40% of intangible-capital payments.

9 Short summary

- The paper measures organization capital using the life cycle of manufacturing plants.
- Organization capital is plant-specific knowledge summarized by productivity and age.
- Free entry gives new plants zero value, but back-loaded rents generate positive aggregate payments.
- The model is calibrated to employment, job creation, and job destruction by plant age.
- The baseline organization-capital share is 3.3% of manufacturing output.
- The incremental contribution is a structural measurement method linking plant dynamics to macro intangible-capital accounting.

10 Conclusion

10.1 Core conclusion

Atkeson and Kehoe show that organization capital is quantitatively important and can be measured using the life cycle of manufacturing plants.

10.2 Economic intuition

Owners finance early low-rent years while plants accumulate operating knowledge. Successful old plants earn back-loaded rents that compensate owners for waiting.

10.3 Incremental contribution takeaway

The paper transforms organization capital from a qualitative concept into a measurable object by combining a plant-dynamics model, U.S. micro plant data, and macro factor-share accounting.

10.4 Limitations

The productivity process is exogenous and reduced-form; worker-owned firm-specific human capital is outside the measurement; and the results are calibrated to manufacturing.

10.5 Final takeaway

Plant growth and survival are signals of an accumulated intangible asset. Organization capital is large, slow to build, risky, and central to understanding owner returns.